

Andrey Goncharov[✉], Nikolai Kondusov, Alexey Zaytsev[✉]LANGUAGE STEERING IN LATENT SPACE TO MITIGATE
UNINTENDED CODE-SWITCHING

Abstract. Multilingual Large Language Models (LLMs) often exhibit hallucinations such as unintended code-switching, reducing reliability in downstream tasks. We propose latent-space language steering, a lightweight inference-time method that identifies language directions via Principal Component Analysis (PCA) on parallel translations and steers token embeddings along these axes to control language identity. Our approach mitigates code-switching while preserving semantics with negligible computational overhead and requires only minimal parallel data for calibration. Empirically, we achieve 95-99% language classification accuracy using a single principal component and reduce next-token distributional divergence by up to 55% across multiple language pairs on Qwen2.5 and Llama-3.2 models. Generation-based evaluation on Llama-3.2 further demonstrates 63–99% reduction in Code-Switching Index across four language pairs ($p < 0.001$). We further analyze the layer-wise evolution of language representations, revealing that language identity concentrates in final layers with near-perfect linear separability.

§1. Introduction

Multilingual large language models (LLMs) frequently exhibit unintended code-switching - generating tokens in languages other than the target despite explicit monolingual instructions. Ryan et al. [13] document how LLM alignment procedures can cause models to default to English even when prompted in other languages, while Yoo et al. [17] demonstrate that code-switching serves as an effective red-teaming vector, bypassing safety filters. In deployed systems, such behavior degrades user trust. For instance, a customer-service chatbot prompted in Spanish may switch to English mid-response, confusing users. Existing solutions require costly fine-tuning or introduce latency via decoding-time interventions.

We propose latent space language steering, a cheap inference-time method that exploits low-dimensional language directions in hidden representations. By applying PCA to parallel translations, we identify linear subspaces separating languages while preserving semantics. Steering via simple

projection (one dot product and vector subtraction per token) mitigates code-switching with negligible overhead.

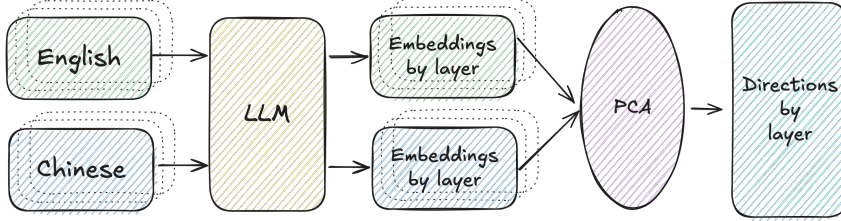


Figure 1. Language steering in hidden space: 1. Collect parallel translations of the same concepts; 2. LLM forward pass to get embeddings by layer; 3. Perform PCA by layer on the merged embeddings.

Our analysis shows language identity concentrates in final layers, with the first principal component (PC1) capturing most of the variance and enabling near-perfect language classification. We validate across Qwen2.5 and Llama-3.2 models on multiple language pairs (English, Spanish, Russian, Chinese, and Hindi), demonstrating reduced distributional divergence and substantially lower code-switching rates in generated text.

Contributions.

- We introduce a cheap steering method with negligible overhead requiring only parallel translations for calibration.
- Empirically validate across models and language pairs, showing reduced code-switching across both distributional and generation-based evaluations.
- Introduce a nearly perfect language classifier based on PCA projections.
- Demonstrate language clustering in multilingual LLMs.
- Code and data are released to facilitate further research ¹.

§2. Related works

Code-switching and multilingual control. While code-switching has been studied for mixed-language inputs in speech and NER [1, 2], recent work

¹<https://github.com/fxlrnpt/language-steering-in-latent-space>

identifies unintended code-switching in multilingual LLM outputs [13, 17], degrading user experience and evaluation. Standard mitigations require costly fine-tuning on monolingual data, reinforcement learning with language-specific rewards, or brittle prompt engineering.

Language structure in representations. Work on mBERT and XLM-R showed language identity concentrates in few principal components, separable from semantics in parallel texts [4, 5]. Yang et al. [16] removed these components for language-agnostic embeddings. Wendler et al. [15] extended this to decoder-only LLMs, revealing latent language preferences. We exploit this structure for active generation control.

Representation steering and concept erasure. Recent work has shown that high-level concepts can be located and manipulated in model representations via linear methods. Zou et al. [18] introduced representation engineering, using PCA on contrastive prompt pairs to identify concept directions (e.g., honesty, harmlessness) and steering activations along them. Turner et al. [14] proposed activation addition, adding steering vectors to the residual stream at inference time. In parallel, concept erasure methods like Ravfogel et al. [12] (iterative null-space projection) and Belrose et al. [3] (closed-form linear erasure) remove specific attributes from representations by projecting onto their null space. Li et al. [9] applied probing-guided inference-time intervention on attention heads to elicit truthful outputs. These methods have primarily targeted sentiment, truthfulness, and toxicity. We apply the same conceptual framework to language identity.

§3. Methods

Our approach operates in the latent space of pre-trained multilingual LLMs without fine-tuning. We identify language-specific directions via PCA on parallel translations, then steer representations along these directions to control code-switching.

3.1. Language Direction Identification. Given a parallel corpus $\mathcal{D} = \{(s_i, k_i)\}_{i=1}^N$ where s_i represents semantically equivalent content in language $k \in \mathcal{K}$, we extract hidden states $\mathbf{h}_i^{(\ell)} \in \mathbb{R}^d$ from each layer ℓ of the LLM. Specifically, we use the hidden state at the last token position of each sentence. For each layer, we pool hidden states across all languages and apply PCA to the centered embeddings $\mathcal{H}^{(\ell)} = \{\mathbf{h}_i^{(\ell)}\}_{i=1}^N$. The first principal component

$\mathbf{v}^{(\ell)}$ captures the language direction:

$$\mathbf{v}^{(\ell)} = \arg \max_{\|\mathbf{v}\|=1} \sum_{i=1}^N \left(\mathbf{v}^\top (\mathbf{h}_i^{(\ell)} - \bar{\mathbf{h}}^{(\ell)}) \right)^2 \quad (1)$$

When semantic content is constant across languages, language identity becomes the dominant variance source, making $\mathbf{v}^{(\ell)}$ an interpretable language axis.

3.2. Latent-Space Language Steering. Empirical analysis (Section 4) shows language variance concentrates in final layers. We apply steering only to layers $\ell \geq \ell_{\text{crit}}$: for Qwen2.5-1.5B (28 layers), we steer layers 27-28; for Llama-3.2-1B (16 layers), we steer layers 15-16. For each token’s hidden state $\mathbf{h}_t^{(\ell)}$ during generation, we modify the language component:

$$\tilde{\mathbf{h}}_t^{(\ell)} = \mathbf{h}_t^{(\ell)} - s \left(\mathbf{h}_t^{(\ell)} \cdot \mathbf{v}^{(\ell)} \right) \mathbf{v}^{(\ell)} \quad (2)$$

where $s \in \mathbb{R}$ controls intervention strength, selected via grid search to optimize either distributional similarity (KL divergence) or generation-based code-switching metrics (Section 4.4). When $s = 1$, this is an exact orthogonal projection that removes the language component. Values $0 < s < 1$ partially attenuate it, while $|s| > 1$ goes beyond projection removal, actively pushing representations away from the source language direction. In practice, optimal coefficients have $|s| > 1$ (see Table 7), indicating that mere removal is insufficient and active counter-steering is needed to overcome the language bias in code-switched input.

3.3. Language Prediction. To validate our approach, we train a logistic regression classifier on final-layer projections $z_i = \mathbf{h}_i^{(L)} \cdot \mathbf{v}^{(L)}$:

$$P(k | z) = \frac{\exp(w_k z + b_k)}{\sum_{k' \in \mathcal{K}} \exp(w_{k'} z + b_{k'})} \quad (3)$$

This one-dimensional classifier predicts language identity with high accuracy, demonstrating the interpretability of discovered directions.

§4. Results

We validate our approach on Qwen2.5-1.5B [11] and Llama-3.2-1B [7] open models across multiple language pairs, demonstrating both the existence of interpretable language directions and their effectiveness for code-switching control.

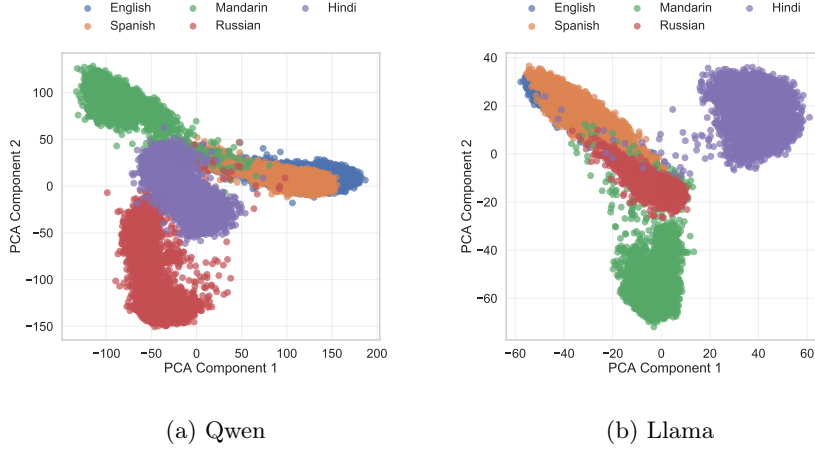


Figure 2. Final layer language map

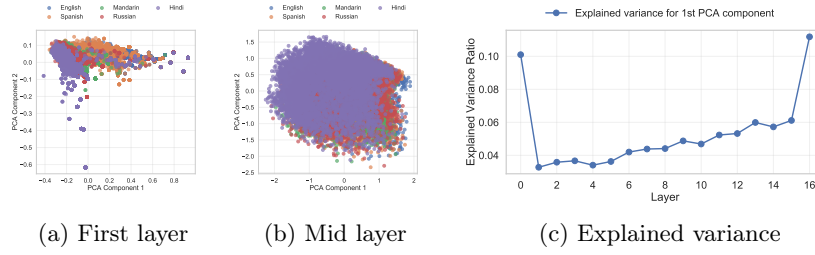
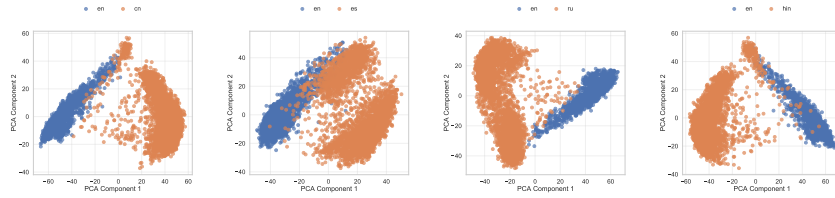


Figure 3. Llama layer-wise language maps and explained variance

4.1. Language direction identification.

Experimental setup. We use Flores Plus [10] dataset with 50 parallel samples (English, Spanish, Russian, Chinese, Hindi) for PCA fitting and 100 for validation.

Language maps reveal emergent clustering. Figure 2a and 2b show final-layer embeddings projected onto the first two PCs. Consistent with prior findings [5, 15], models organize identical content into tight, language-specific clusters with clear boundaries. Languages within the Indo-European family (Spanish, Russian) cluster closer than typologically distant pairs



(a) English - Chinese (b) English - Spanish (c) English - Russian (d) English - Hindi

Figure 4. Llama final layer language maps for each language pair

(English–Chinese), suggesting PC1 captures coarse language distinction while PC2 encodes finer linguistic properties.

Figures 3a–3b show layer-wise evolution: clusters emerge loosely in early layers and sharpen dramatically in final layers. Figure 3c reveals late-stage specialization for language identity. Pairwise maps (Figures 4a, 4b, 4c, 4d) confirm clean linear separability for all tested pairs.

4.2. Language prediction.

Experimental setup. We use Flores Plus [10] dataset with 50 parallel samples (English, Spanish, Russian, Chinese, Hindi) for PCA and logistic regression fitting and 100 for validation.

Classifier shows nearly perfect performance. Table 1 demonstrates that a simple logistic regression classifier trained only on the first principal component achieves near-perfect accuracy (0.96–1.00) across all language pairs and both models. This validates that language identity information is linearly accessible in the final layer representations, supporting our hypothesis that models maintain explicit language directions in their activation space. We observe decreased performance for the English–Spanish pair in both models, attributable to these languages being more closely related to each other.

4.3. Language steering.

Experimental setup. We fit the steering module on Flores Plus with 200 parallel samples and test on 2,467 TED Talks samples with artificial code-switching. Each sample is truncated to 20 words, split at the midpoint, and the second half is translated to the target language using Gemini 2.5

Language pair	Qwen	Llama
English - Chinese	0.98	0.98
English - Spanish	0.96	0.96
English - Russian	0.99	0.99
English - Hindi	0.98	1.00

Table 1. Last layer language classifier accuracy

Flash via the OpenRouter API, producing texts that begin in English and switch mid-sentence. We note that this artificial code-switching created by translating the second half of each sentence may not capture the full complexity of naturally occurring code-switching. We use this controlled setup to isolate the effect of steering but acknowledge the need for evaluation on naturally occurring code-switched text in future work. Success is measured by comparing the next-token prediction distribution of the steered model against the English baseline using four complementary metrics: KL divergence $KL(P_{\text{EN}} \| P_{\text{steered}})$, Jensen-Shannon divergence, cosine similarity, and top-50 token overlap. Lower divergence and higher similarity/overlap indicate better mitigation.

To find the best steering strength coefficient we conduct a grid search with the best values listed in Table 7 for each language pair.

Results. Table 2 demonstrates that steering shifts predictions from Spanish tokens (muy, bastante, tan) to their English equivalents (very, quite, so) while preserving semantic coherence. For well-calibrated coefficients, steering changes how the model speaks rather than what it says, though aggressive coefficients can disrupt semantic content (see Limitations).

Table 3 shows steering reduces KL divergence by 25-55% across language pairs and models (38% average for Qwen, 43% for Llama). While the method substantially mitigates code-switching, the remaining gap indicates that the complex structure of the manifold does not allow full reconstruction of the original concept with a simple linear transformation. The method shows strongest performance for typologically distant pairs: English-Chinese achieves 55% KL reduction on Qwen, while English-Russian achieves 55% on Llama. We hypothesize that model-specific tokenization and training data balance play a key role: the Qwen model processed much more English and Chinese data, leading to more granular tokenization and better internal generalization for that pair. English-Hindi remains the weakest pair for both models (24-27% reduction).

Original	Unsteered	Steered
still	muy	very
going	un	surprised
not	bastante	pretty
so	tan	starting
a	sor	working
just	mΓŸs	content
really	a	rather
in	empez	impressed
very	de	quite
sitting	content	—
quite	en	great
pretty	enc	beginning
here	aquΓ	answering
thinking	pens	almost
looking	realmente	fully
kind	dese	— — — —
getting	completamente	— —
having	feliz	even
actually	org	heart
surprised	seguro	trying

Table 2. Top 20 token sample (Qwen, English-Spanish)

These findings are supported by supplementary metrics. JS divergence (Table 4) shows consistent improvement across all pairs. Cosine similarity between steered and English baseline distributions increases substantially (Table 5), e.g., from 0.02 to 0.22 for Qwen English-Chinese. Top-50 token overlap (Table 6) reveals that steering dramatically increases the number of shared high-probability tokens with the English baseline - from under 8% to over 25% for most pairs. English-Hindi on Qwen again shows minimal improvement across all supplementary metrics (cosine: 0.04 $\rightarrow$ 0.07, overlap: 11.23% $\rightarrow$ 11.68%), reinforcing that this is the most challenging pair.

Notably, the optimal steering coefficient for Qwen English-Hindi is positive (+5.0), indicating a reversed PCA direction for this pair relative to all others. This may reflect tokenization asymmetries in the Qwen model’s treatment of Devanagari script. In contrast, Llama uses a consistent negative direction across all language pairs (Table 7).

Language pair	Qwen	Llama
English - Chinese	10.15 $\rightarrow$ 4.52	9.05 $\rightarrow$ 4.77
English - Spanish	7.25 $\rightarrow$ 4.90	6.52 $\rightarrow$ 3.48
English - Russian	8.70 $\rightarrow$ 5.40	7.77 $\rightarrow$ 3.49
English - Hindi	9.60 $\rightarrow$ 7.02	8.06 $\rightarrow$ 6.13

Table 3. KL divergence (no steering $\rightarrow$ steering)

Language pair	Qwen	Llama
English - Chinese	0.67 $\rightarrow$ 0.49	0.65 $\rightarrow$ 0.51
English - Spanish	0.59 $\rightarrow$ 0.52	0.58 $\rightarrow$ 0.44
English - Russian	0.62 $\rightarrow$ 0.50	0.62 $\rightarrow$ 0.43
English - Hindi	0.64 $\rightarrow$ 0.60	0.63 $\rightarrow$ 0.54

Table 4. JS divergence (no steering $\rightarrow$ steering)

Language pair	Qwen	Llama
English - Chinese	0.02 $\rightarrow$ 0.22	0.05 $\rightarrow$ 0.21
English - Spanish	0.12 $\rightarrow$ 0.18	0.12 $\rightarrow$ 0.29
English - Russian	0.09 $\rightarrow$ 0.23	0.10 $\rightarrow$ 0.31
English - Hindi	0.04 $\rightarrow$ 0.07	0.06 $\rightarrow$ 0.16

Table 5. Cosine similarity (no steering $\rightarrow$ steering, higher = better)

Language pair	Qwen	Llama
English - Chinese	3.69% $\rightarrow$ 26.46%	4.47% $\rightarrow$ 24.31%
English - Spanish	11.41% $\rightarrow$ 20.23%	11.45% $\rightarrow$ 30.74%
English - Russian	7.41% $\rightarrow$ 25.45%	7.49% $\rightarrow$ 31.96%
English - Hindi	11.23% $\rightarrow$ 11.68%	6.54% $\rightarrow$ 25.88%

Table 6. Top-50 token overlap (no steering $\rightarrow$ steering, higher = better)

4.4. Generation-based evaluation. To complement the distributional analysis above, we evaluate steering on actual text generation using Llama-3.2-1B. For each of 500 TED Talks samples per language pair, we generate 100-token continuations in three conditions: (A) English-only input without

Language pair	Qwen	Llama
English - Chinese	-3.2	-5.0
English - Spanish	-2.8	-3.7
English - Russian	-2.2	-3.5
English - Hindi	+5.0	-3.6

Table 7. Best steering strength coefficient (KL-optimized)

steering, (B) code-switched input without steering, and (C) code-switched input with steering. We measure code-switching in the generated output using standard metrics [6, 8]: Code-Switching Index (CSI), the fraction of tokens not in the target language (lower = better); M-Index, a multilingual diversity measure (0 = monolingual, 1 = maximally mixed); I-Index, the ratio of language switch points to token boundaries; and Target Language Consistency (TLC = $1 - \text{CSI}$, higher = better). Token-level language identification uses a FastText classifier with a sliding window of 5 tokens. Steering coefficients. We evaluate two sets of steering coefficients (Table 8). The KL-optimized set uses the coefficients from the distributional grid search (Table 7). The CSI-optimized set is obtained by a separate grid search that directly minimizes CSI on a 10-sample calibration subset. CSI-optimized coefficients tend toward higher magnitudes, which improves language control but increases degeneration risk (see Limitations).

Language pair	KL-opt.	CSI-opt.
English - Chinese	-5.0	-5.0
English - Spanish	-3.7	-5.0
English - Russian	-3.5	-4.0
English - Hindi	-3.6	-5.0

Table 8. Llama-3.2-1B steering coefficients: KL-optimized (from distributional analysis) vs. CSI-optimized (from generation grid search).

Results. Table 9 shows that steering substantially reduces code-switching in generated text. With KL-optimized coefficients, CSI drops by 63–95%

across language pairs, demonstrating that distributional optimization transfers well to generation. CSI-optimization yields further gains for English-Spanish ($0.23 \rightarrow 0.11$) and English-Hindi ($0.04 \rightarrow 0.01$), but at the cost of higher-magnitude coefficients that increase degeneration risk.

Language pair	KL-optimized		CSI-optimized	
	CSI ($\downarrow$)	TLC ($\uparrow$)	CSI ($\downarrow$)	TLC ($\uparrow$)
Unsteered (code-switched input, no steering):				
English - Chinese	0.64 / 0.36		—	
English - Spanish	0.62 / 0.38		—	
English - Russian	0.59 / 0.41		—	
English - Hindi	0.81 / 0.19		—	
Steered:				
English - Chinese	0.03	0.97	0.03	0.97
English - Spanish	0.23	0.77	0.11	0.89
English - Russian	0.02	0.98	0.01	0.99
English - Hindi	0.04	0.96	0.01	0.99

Table 9. Generation-based code-switching metrics (Llama-3.2-1B, $n = 500$). English-only baseline: CSI = 0.02. All steered vs. unsteered differences significant at $p < 0.001$ (Wilcoxon signed-rank test).

Notably, steering coefficients are highly fragile: small changes in magnitude can shift the model from fluent code-switched output to degenerate repetition. The CSI-optimized coefficients cluster near the maximum tested value (-5.0), suggesting that direct optimization of generation metrics pushes toward an aggressive regime where language control and fluency trade off sharply.

Qualitative analysis. Table 10 shows representative examples. Steering successfully redirects generation from the foreign language to English, but aggressive coefficients can produce repetitive output patterns, indicating a trade-off between language control and generation fluency that we discuss further in Limitations.

§5. Conclusion

We introduced latent-space language steering, a lightweight inference-time method that mitigates unintended code-switching in multilingual LLMs without fine-tuning. By extracting language directions via PCA on

en-es: "...I took on the task to enseñar desarrollo global..."	
Unsteered	de que un grupo de estudiantes suecos, que habían sido enviados a la Universidad de Harvard, me escribieron para pedir ayuda...
Steered	de that I had the opportunity to work with the first two editions of the course and I was very inspired by it...
en-ru: "...About 10 years ago, I took on the task to [Russian text]..."	
Degenerate steered	first year of the first year of the first year of the first year of the first year...

Table 10. Generated continuations (Llama-3.2-1B, KL-optimized). Top: steering successfully redirects Spanish to coherent English. Bottom: steering eliminates Russian but produces degenerate repetition.

parallel translations, our approach enables precise control over language identity with negligible overhead - one dot product and vector subtraction per token.

Our empirical analysis across Qwen2.5 and Llama-3.2 models reveals that language identity concentrates in final layers with near-perfect linear separability. This geometric structure enables both accurate language classification (95-99% accuracy) and effective steering that reduces distributional divergence by up to 55% and achieves 63-99% Code-Switching Index reduction in generated text. The layer-wise evolution shows language identity crystallizes in final layers while early layers focus on semantics, suggesting these occupy largely orthogonal subspaces.

While distributional divergence is reduced by 40% on average and generated text shows near-complete language conversion ($TLC > 0.77$ across all pairs), steering coefficients are highly fragile - aggressive values eliminate code-switching but can produce degenerate repetition, revealing a sharp trade-off between language control and generation fluency. Our approach demonstrates that interpretable control mechanisms can be extracted from learned representations, offering a promising direction toward lightweight alternatives to fine-tuning for multilingual language control, though addressing the fluency-control trade-off and coefficient sensitivity remains important future work. Natural extensions include multilingual steering

(controlling language choice among three or more languages simultaneously) and bidirectional steering (e.g., steering from English toward Spanish), both of which the geometric structure of the multilingual PCA maps suggests should be feasible.

Limitations

- Model scale and diversity. Experiments limited to small models (1-1.5B parameters) and two architectures (Qwen2.5, Llama-3.2). Larger models and diverse architectures remain unexplored.
- Incomplete reconstruction. While the method achieves up to 55% KL divergence reduction, the remaining gap indicates linear projections cannot fully reconstruct monolingual distributions, suggesting non-linear language-semantic interactions at the manifold level.
- Dataset coverage. Evaluation relies on Flores Plus and TED Talks (formal text). Conversational data, technical domains, and naturally occurring code-switching patterns are not tested. Our artificial code-switching setup produces mid-sentence switches at fixed positions, which may not reflect the morphosyntactic boundaries where natural code-switching typically occurs. Evaluation on naturalistic code-switching corpora such as the LinCE benchmark [1] is an important direction for future work.
- Language typology. Limited to Indo-European and Chinese languages. Low-resource languages, right-to-left scripts, and typologically distant languages (agglutinative, tonal) unexplored. Hindi showed the weakest improvement across both models, with KL reduction (24-27%) but minimal gains in other distributional similarity metrics, suggesting tokenization or data balance issues.
- Layer-wise steering. Steering applied only to final layers based on empirical observation. Systematic exploration of multi-layer or adaptive steering strategies not conducted.
- Evaluation metrics. While we supplement distributional metrics (KL, JS divergence) with generation-based code-switching metrics (CSI, M-Index, I-Index), we do not evaluate semantic preservation or downstream task performance. Human evaluation of output quality remains needed.

- Fluency–control trade-off and coefficient fragility. Generation-based evaluation reveals that aggressive steering coefficients, while effective at eliminating code-switching ($\text{CSI} < 0.01$ for English-Russian and English-Hindi), can produce repetitive or degenerate output patterns. Steering coefficients are highly fragile. Small changes in magnitude shift behavior from fluent code-switched text to degenerate repetition. The CSI-optimized coefficients cluster near the maximum tested values (-5.0), suggesting that direct optimization of generation metrics pushes toward an aggressive regime. Combining steering with constrained decoding or adaptive coefficient scheduling could mitigate this issue.
- Steering calibration. Optimal coefficients require manual grid search per language pair and depend on the optimization target (distributional vs. generation-based). Principled or adaptive selection methods not developed.
- Efficiency benchmarking. No rigorous latency profiling across hardware configurations or batch sizes to validate overhead claims for production deployment.
- Bilingual, English-only steering. Our evaluation steers exclusively toward English in bilingual settings. Multilingual steering (controlling language choice among three or more languages simultaneously) and bidirectional steering (e.g., steering from English toward Spanish) are natural extensions that we leave to future work. The multilingual PCA maps (Figures 2a, 2b) suggest the geometric structure supports such extensions, as multiple language clusters are simultaneously separable.

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